FALCON MODELS
BUSINESS ANALYSIS PROJECT

-KASHISH PAL

PROJECT STRUCTURE

PROJECT OVERVIEW

Falcon Models is a global scale model retailer selling across 19 countries. This project analyses three years of sales, customer, payment, and product data to identify what is driving performance, where the business is exposed to risk, and what actions can improve revenue stability and cash flow.

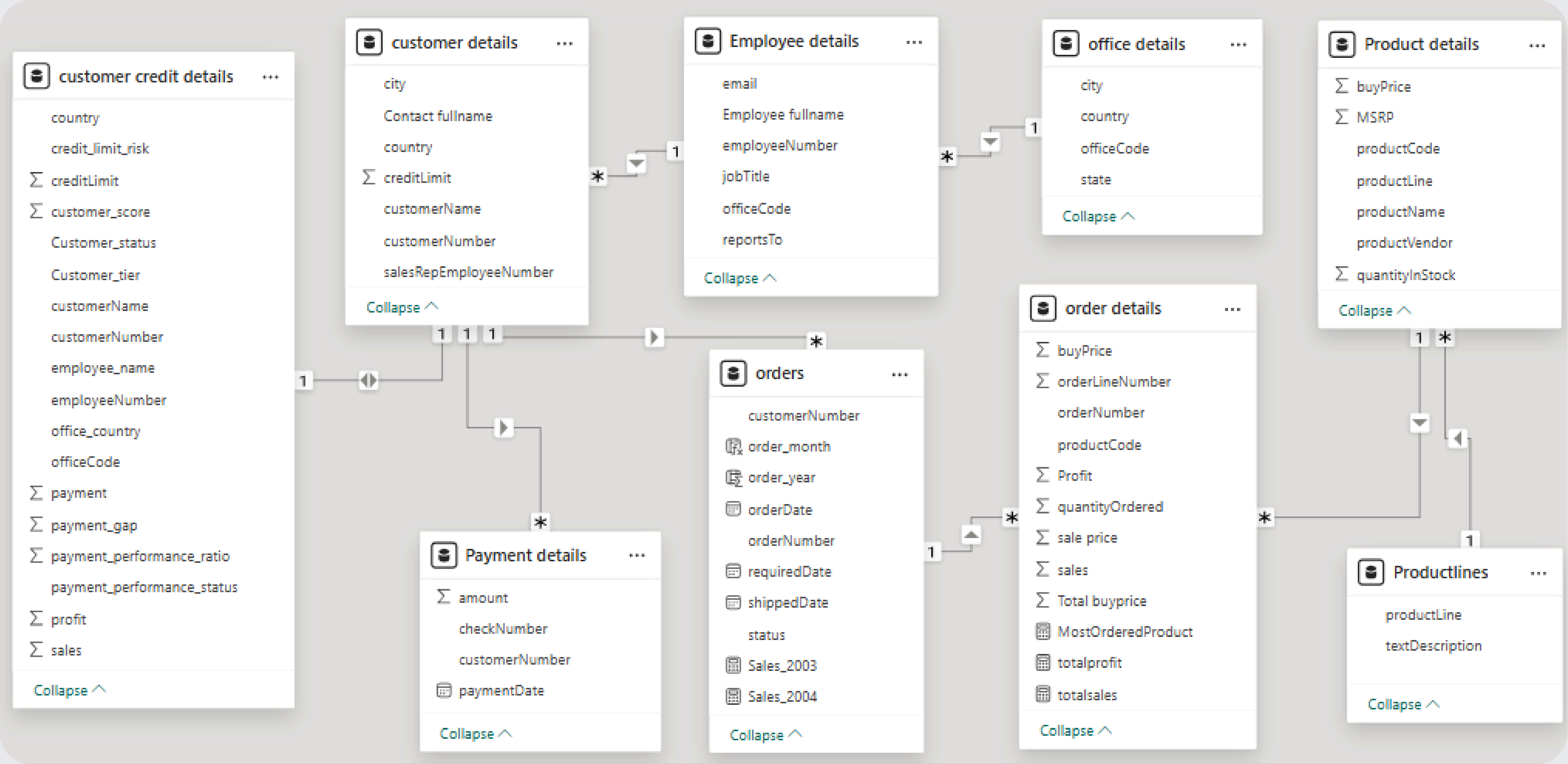
BUSINESS PROBLEM

Revenue depends heavily on a small number of customers, employees, and markets. Payment collection is inconsistent, with significant outstanding balances sitting unresolved. There is no clear view of which customers are financially reliable, which product lines are worth investing in, and where the next growth opportunity lies.

Approach:

- Built a 5-CTE SQL pipeline across 7 tables to engineer payment performance ratios, credit risk flags, and composite customer scores, segmenting customers into VIP, Premium, and Regular tiers
- Built a 3-page interactive Power BI dashboard covering sales performance, product profitability, and customer credit health
- Delivered quantified insights and recommendations directly tied to business risk and revenue opportunity

DATA MODEL OVERVIEW



FALCON MODELS SALES & CUSTOMERS ANALYSIS

- Welcome to the Falcon Models executive dashboard. This report is designed to give a complete view of sales performance, product trends, customer credit insights, and employee contributions.
- Below is an at-a-glance overview of the dashboard structure. Use the navigation buttons to explore each section in detail.

Executive Summary | *Product Performance Analysis* | *Customer Credit & Risk Analysis*

Page 1 : Executive Summary

- How much did the business grow year on year and what does 2005 look like?
- Which product lines are generating the most profit?
- How dependent is the business on its top employees for revenue?
- How is revenue distributed across global markets?

Revenue performance, profit, growth & employee contribution-the full business picture in one view,

Page 2: Product Performance Analysis

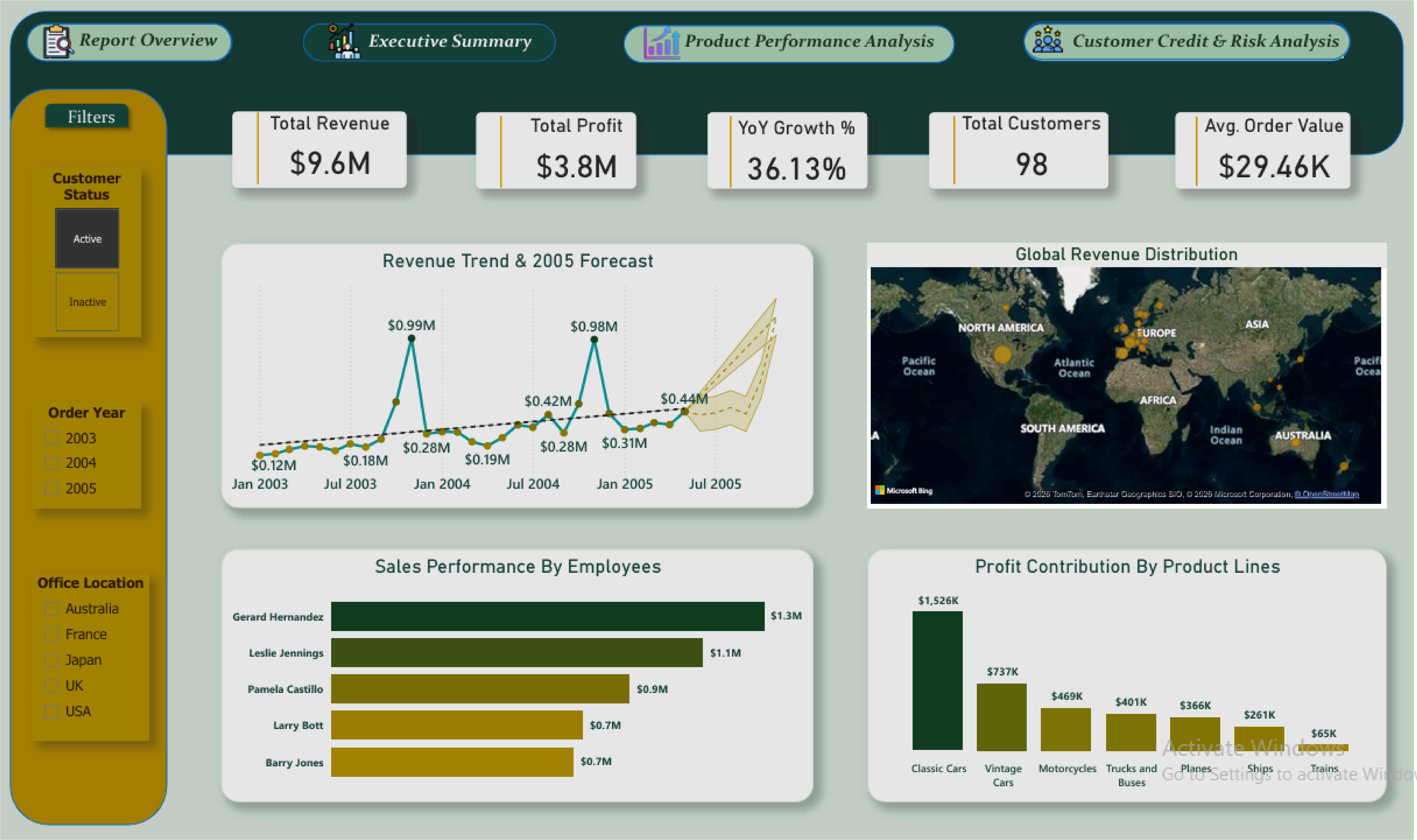
- Which products and product lines generate the most revenue and profit?
- Which product line has the highest margin but the lowest sales attention?
- Which markets are driving sales and which are underperforming their potential?
- Are the bottom performers worth keeping or should they be phased out?

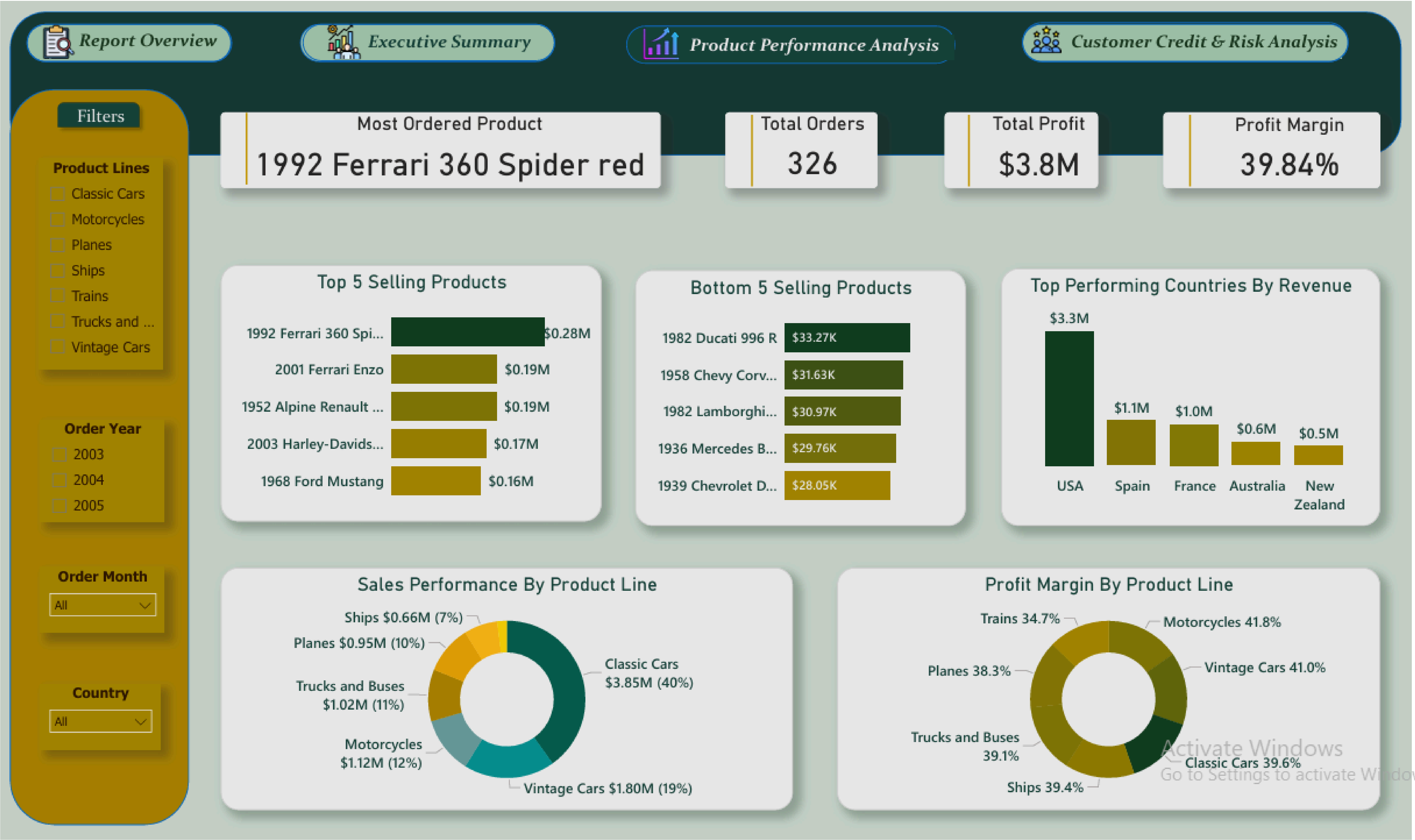
What's selling, what's not, and where the real profit margin sits.

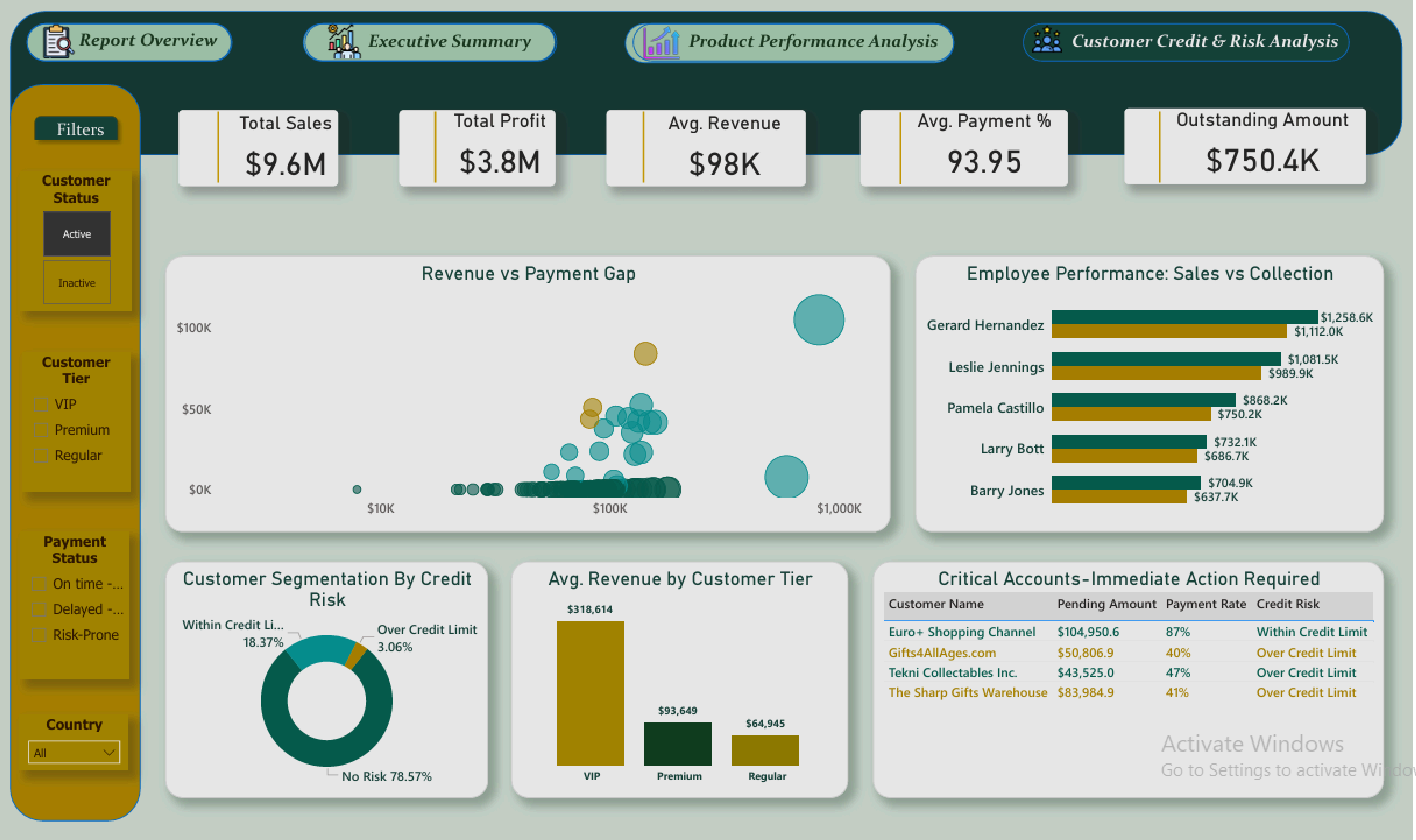
Page 3: Customer Credit & Risk Analysis

- Which customers are reliable and which are a financial risk?
- How much revenue is sitting uncollected and who owes it?
- Are high-value customers also the most financially stable?
- Are top revenue generating employees also collecting what they sell or have payment collection gaps?

A clear view of who is valuable, who is financially risky, and where $750K in outstanding payments is sitting.


Report Overview
Executive Summary
Product Performance Analysis
Customer Credit & Risk Analysis
Filters
Customer Status
Active
Inactive
Order Year
2003
2004
2005
Office Location
Australia
France
Japan
UK
USA
Total Revenue
$9.6M
Total Profit
$3.8M
YoY Growth %
36.13%
Total Customers
98
Avg. Order Value
$29.46K
Revenue Trend & 2005 Forecast
$0.99M
$0.98M
$0.42M
$0.44M
$0.12M
$0.18M
$0.28M
$0.19M
$0.28M
$0.31M
Jan 2003
Jul 2003
Jan 2004
Jul 2004
Jan 2005
Jul 2005
Global Revenue Distribution
NORTH AMERICA
EUROPE
ASIA
Pacific Ocean
Atlantic Ocean
AFRICA
SOUTH AMERICA
Indian Ocean
AUSTRALIA
Microsoft Bing
Sales Performance By Employees
Gerard Hernandez
$1.3M
Leslie Jennings
$1.1M
Pamela Castillo
$0.9M
Larry Bott
$0.7M
Barry Jones
$0.7M
Profit Contribution By Product Lines
$1,526K
Classic Cars
$737K
Vintage Cars
$469K
Motorcycles
$401K
Trucks and Buses
$366K
Planes
$261K
Ships
$65K
Trains
Activate Windows
Go to Settings to activate Windows


Report Overview
Executive Summary
Product Performance Analysis
Customer Credit & Risk Analysis
Filters
Product Lines
Classic Cars
Motorcycles
Planes
Ships
Trains
Trucks and ...
Vintage Cars
Order Year
2003
2004
2005
Order Month
All
Country
All
Most Ordered Product
1992 Ferrari 360 Spider red
Total Orders
326
Total Profit
$3.8M
Profit Margin
39.84%
Top 5 Selling Products
1992 Ferrari 360 Spi... $0.28M
2001 Ferrari Enzo $0.19M
1952 Alpine Renault ... $0.19M
2003 Harley-Davids... $0.17M
1968 Ford Mustang $0.16M
Bottom 5 Selling Products
1982 Ducati 996 R $33.27K
1958 Chevy Corv... $31.63K
1982 Lamborghi... $30.97K
1936 Mercedes B... $29.76K
1939 Chevrolet D... $28.05K
Top Performing Countries By Revenue
USA $3.3M
Spain $1.1M
France $1.0M
Australia $0.6M
New Zealand $0.5M
Sales Performance By Product Line
Ships $0.66M (7%)
Planes $0.95M (10%)
Trucks and Buses $1.02M (11%)
Motorcycles $1.12M (12%)
Vintage Cars $1.80M (19%)
Classic Cars $3.85M (40%)
Profit Margin By Product Line
Trains 34.7%
Motorcycles 41.8%
Planes 38.3%
Vintage Cars 41.0%
Trucks and Buses 39.1%
Classic Cars 39.6%
Ships 39.4%
Activate Windows
Go to Settings to activate Windows.


Report Overview
Executive Summary
Product Performance Analysis
Customer Credit & Risk Analysis
Filters
Customer Status
Active
Inactive
Customer Tier
VIP
Premium
Regular
Payment Status
On time -...
Delayed -...
Risk-Prone
Country
All
Total Sales
$9.6M
Total Profit
$3.8M
Avg. Revenue
$98K
Avg. Payment %
93.95
Outstanding Amount
$750.4K
Revenue vs Payment Gap
$100K
$50K
$0K
$10K
$100K
$1,000K
Employee Performance: Sales vs Collection
Gerard Hernandez $1,258.6K $1,112.0K
Leslie Jennings $1,081.5K $989.9K
Pamela Castillo $868.2K $750.2K
Larry Bott $732.1K $686.7K
Barry Jones $704.9K $637.7K
Customer Segmentation By Credit Risk
Within Credit Li... 18.37%
Over Credit Limit 3.06%
No Risk 78.57%
Avg. Revenue by Customer Tier
$318,614
$93,649
$64,945
VIP
Premium
Regular
Critical Accounts-Immediate Action Required
Customer Name | Pending Amount | Payment Rate | Credit Risk
Euro+ Shopping Channel | $104,950.6 | 87% | Within Credit Limit
Gifts4AllAges.com | $50,806.9 | 40% | Over Credit Limit
Tekni Collectables Inc. | $43,525.0 | 47% | Over Credit Limit
The Sharp Gifts Warehouse | $83,984.9 | 41% | Over Credit Limit
Activate Windows
Go to Settings to activate Windows

DAX MEASURES & CALCULATIONS

Advanced measures built for segmentation, profitability, and performance analysis. Other metrics used for analysis are Total Sales, Total Profit, Total Customers, Total Orders, Most Ordered Product, Avg. Shipping Days & Outstanding Amount.

Avg. Order Value: $ 29.46K Revenue earned per order on average.

AOV = DIVIDE([totalsales], DISTINCTCOUNT('order details'[orderNumber]))

YoY Revenue Growth : **36.3%** Revenue growth from 2003 to 2004.

YoY Growth % = DIVIDE([Sales_2004] - [Sales_2003], [Sales_2003])

Avg. Revenue per Customer: $78.7 K Total revenue divided by total customers. Used to compare spending across VIP, Premium, and Regular tiers.

Avg. Revenue per Customer = DIVIDE([totalsales], DISTINCTCOUNT('customer credit details'[customerNumber]))

Most Ordered Product: 1992 Ferrari 360 Spider Red Dynamically identifies the product with the highest quantity ordered.

MostOrderedProduct = SELECTCOLUMNS(TOPN(1, SUMMARIZE('Product details', 'Product details'[productName], "Totalordered", SUM('order details'[quantityOrdered])), [Totalordered], DESC), "MostOrderedProduct", 'Product details'[productName])

Profit Margin %: **39.7%** Profit as a percentage of sales. Used across product lines to identify where the business earns most per sale.

Profit Margin % = DIVIDE([totalprofit], [totalsales])

INSIGHTS REPORT

Over the last three years, the business has shown healthy growth, but the data also reveals some critical dependencies and opportunities for diversification. Our revenue is heavily concentrated in specific markets, product lines, and customer groups: strengths that drive stability but also create risk if not balanced.

1. Growth is strong but concentrated- 2005 tracking ahead

Revenue grew 36% from $3.3M in 2003 to $4.5M in 2004. Jan–May 2005 alone has already generated $1.79M-tracking ahead of both prior years at the same point. Based on the two-year seasonal pattern, 2005 is projected to close between $5.1M-$5.4M. However, this growth rests on just 3 countries (USA, Spain & France) driving 56% of sales and 7 customers generating 23.2% of revenue.

2003 → 2004: +36% || Jan–May 2025: $1.79M || 2005 projected: $5.1M–$5.4M || 3 countries = 56% of sales

2. Classic Cars lead sales-Motorcycles lead margins

Classic and Vintage Cars are the core of the business, together accounting for nearly 60% of all profit. The top selling product: the 1992 Ferrari 360 Spider Red alone generated $276,840. However, Motorcycles quietly deliver the highest profit margin in the entire productline at 40.7%, yet make up only 11.7% of sales.

Classic + Vintage = 59.2% of profit || Top product: Ferrari 360 Spider: $276,840 || Motorcycles: highest margin at 40.7%

3. Trains form the weakest line

At just $188,533 in revenue and a 34.6% margin, Trains are the lowest performer across both sales and profit. Trains appeal to casual buyers and children's gifting, while every other product line serves premium collectors: classic car enthusiasts, motorcycle fans, aviation buyers. The line needs a rethink or gradual phase out.

Trains: $188,533 revenue (2% of sales) || Lowest margin: 34.6%

4. The top customer carries the highest unpaid balance

Euro+ Shopping Channel is the single largest customer at $820,689 in revenue, yet carries $104,951 in unpaid balance and is the only VIP flagged as a delayed payer. All 6 other VIP customers have a perfect, zero-outstanding payment record.

VIP avg revenue: $318,614 || Euro+ revenue: $820,689 || Euro+ unpaid: $104,951

5.$750K in sales remains uncollected

13 customers have paid less than 75% of what they owe, with a combined outstanding balance of $565K. Three of them have already exceeded their credit limits: Tekni Collectables, Gifts4AllAges, and The Sharp Gifts Warehouse. Another 6 are approaching the same threshold with $175K outstanding between them.

Total outstanding: $750K || High-risk pool: 13 customers, $565K || Over credit limit: 3 customers

6. Three employees carry one-third of revenue

The top 3 employees bring in $3.52M (33.4% of total sales). The bottom 5 manage a similar number of accounts but generate only 21.5%. If a top performer leaves, there is currently no plan or shared knowledge base to absorb the impact.

Top 3 = $3.52M (33.4% of sales) || Top 5 = 48.3% of sales || Bottom 5 = 21.5% of sales

7. November drives more than any other month

November averaged $983,658 across 2003 and 2004: the highest month both years. Q4 as a whole contributes 38.4% of annual revenue, while the rest of the year averages just $328,708 per month. The business runs on one seasonal peak.

November avg: $983,658 || Q4 = 38.4% of annual revenue || Non Q4 monthly avg: $328,708

8. Spain ranks 2nd globally in revenue - with no local office

At $1.06M and 11% of total sales, Spain outperforms France, UK, and Japan despite having no dedicated office. It is entirely remotely serviced and has significant room to grow further with structured local support.

Spain revenue: $1.06M (11% of sales) || Ranked 2nd globally || Outperforms 3 office locations

RECOMMENDATIONS

- **Euro+ Shopping Channel: Address $105K outstanding now:** Euro+ is our highest revenue customer at $820,689, yet the only VIP carrying $104,951 in unpaid balance with a Delayed payment status. Assign dedicated account management, initiate a structured payment plan, and offer an early settlement discount. Losing this account would require 13 new Regular customers to compensate.

- **Re-engage Inactive customers:** 24 customers have zero orders and a $0 credit limit, they were onboarded but never converted. A targeted first order offer or seasonal campaign could activate even 5–7 of them, generating $325K–$455K at the Regular tier average.

- **VIP retention: Protect 23% of revenue from churn:** 7 VIP customers generate $2.23M (23.2% of revenue) at an average of $318,614 each. Five of seven have zero outstanding balances and 100% payment reliability. Introduce quarterly account reviews and ensure no VIP relationship is managed by a single salesperson with no backup.

- **Motorcycles: Highest margin line, lowest promotion:** Motorcycles carry the highest profit margin at 40.7% but only 11.7% of total sales. Bundle Motorcycle models with top selling Classic Cars in promotions. This lifts the highest-margin line without needing to build a new audience from scratch.

- **Credit risk: Freeze 3 accounts, recover up to $339K:** Tekni Collectables, Gifts4AllAges, and The Sharp Gifts Warehouse have exceeded their credit limits, place them on order hold immediately. For the broader 13 Risk-Prone customers ($565K outstanding), implement automated 30-60-90 day reminders and offer a 1.5% early-payment discount. Collecting 40–60% of this pool recovers $226K–$339K in cash without a single new sale.

- **Sales team: Protect revenue if a top performer leaves:** The top 3 salespeople bring in $3.52M, one-third of all company revenue. If any one of them leaves, that revenue is at immediate risk. The bottom 5 salespeople handle a similar number of accounts but generate only 21.5% of sales. Sharing best practices from top sales performers, implementing mentorship programs, and introducing team-based targets can help improve overall sales efficiency and collaboration.

- **Seasonality: Build off-peak revenue, reduce Q4 dependence:** Non Q4 months average is just $328,708 vs $614,575 in Q4, a 53% drop. November alone averages $983K across both years. Mid-year collector launches, anniversary promotions, and gifting campaigns timed for Q2–Q3 could add approximately $296K annually at a 10% improvement in off-peak monthly revenue.

PROJECT SUMMARY

The Falcon Models analysis gives a clear picture of sales, customer payments, and product performance. Key findings show steady sales trends with seasonal peaks, top products driving revenue, and differences in how customers handle payments. These insights highlight both growth opportunities and areas to manage financial risk.

This project combines structured data modeling with interactive dashboards, ensuring that business leaders can not only track performance but also drill deeper into underlying drivers. By integrating SQL-based data preparation with Power BI visuals, the analysis translates raw data into actionable insights for informed decision-making.

Thankyou for reviewing this project!
-KASHISH PAL

CONTACT INFO:

: Lets connect on linkedin!   : Kashishpal2020@gmail.com   : View my portfolio!